

Software Requirements Specification (SRS) Documentation

Project: Event Management System(Version 1.0)

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Team	SRS for Event Management System;	
1.0	06-09-2026	Gokul	Introduction, Overall description, System features	
1.0	06-09-2026	Hitha	acceptance tests, security requirements and acceptance tests.	
1.0	06-09-2026	Jagriti	Functional and Non functional requirements listed, external interface	
1.0	06-09-2026	Aaryesh	External interface requirements, UMS diagrams and Requirements traceability matrix	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for an Event Management System. It defines the functional and non-functional requirements, interfaces, security requirements, and verification criteria for managing events and event registrations.

1.2 Scope

The system covers user registration and authentication, event creation and management, event discovery, event registration, payment handling, ticket generation, participant management, notifications, and administrative monitoring. It focuses on the event management and registration functions of the system.

1.3 Audience

Developers, QA Engineers, System Administrators, Event Organizers, Users, and Assessment Evaluators.

1.4 Definitions

List of acronyms: SRS, UI, API, OTP, QR, DB, HTTPS, Admin.

2. Overall description

2.1 Product perspective

The Event Management System is a web-based application that allows users to discover events and register for them, while event organizers can create and manage events and participants. The system uses a web user interface, application services, a database, and external payment and notification services where required.

2.2 Major product functions (detailed)

- User registration and login
- Event creation, update, and deletion
- Event search and filtering
- Event details display
- Event registration
- Registration confirmation and ticket generation
- Online payment processing
- Registration cancellation
- Participant management
- Event capacity management
- Notifications and reminders
- Event reports
- User management
- Event approval and disabling
- System monitoring and audit logging

2.3 User roles and characteristics (expanded)

- User: can create an account, browse events, register for events, make payments, view tickets, and manage registrations.
- Event Organizer: can create and manage events, configure capacity, view registrations, manage participants, and generate event reports.
- Administrator: can manage users, approve or disable events, monitor system activity, and review audit information.

2.4 Operating environment

The system operates as a web application accessible through modern web browsers. The application uses a server-side application environment, a database server, and network connectivity for communication with external services.

2.5 Constraints

- The system shall require network connectivity for online access and payment processing.
- Access to protected functions shall require authenticated users with appropriate roles.
- Payment processing shall depend on the availability of the configured payment gateway.
- The system shall protect user and registration data from unauthorized access.

3. External interface requirements

3.1 User interfaces

The primary UI shall be a responsive web interface with clear navigation for login, event discovery, event details, registration, payment, tickets, organizer functions, and administrator functions. Forms shall provide clear validation and error messages.

3.2 Hardware interfaces

- Computer, laptop, tablet, or smartphone for accessing the web application.
- Display and input devices required to interact with the web interface.

3.3 Software interfaces

- Database interface for storing users, events, registrations, payments, tickets, and audit information.
- Payment Gateway API for online payment processing.
- Email or notification service for registration confirmations and event notifications.

3.4 Communications

- The system shall use HTTPS for communication between the client and server.
- API requests and responses shall use structured data formats such as JSON where applicable.

- The system shall handle communication failures and display an appropriate status to the user.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow EMS-F-####.

4.1 User account and authentication

Requirements for account creation, login, and profile management.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-001	The system shall allow a user to create an account using required registration details.	Functional	High	User	AC-EMS-F-001: Valid details create an account and display confirmation. Test: TC-REG-01	Database required
EMS-F-002	The system shall authenticate registered users using valid login credentials.	Functional	High	User	AC-EMS-F-002: Correct credentials open the appropriate user session. Test: TC-AUTH-01	User account required
EMS-F-003	The system shall allow an authenticated user to update permitted profile information.	Functional	Medium	User	AC-EMS-F-003: Updated valid profile data is saved and displayed. Test: TC-PROF-01	Authentication required

4.2 Event management

Requirements for organizers to create, update, delete, search, and display event information.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-004	The system shall allow organizers to create an event with title, description, date, time, venue, capacity, and registration details.	Functional	High	Event Organizer	AC-EMS-F-004: Valid event details create an event record. Test: TC-EVT-01	Organizer role required
EMS-F-005	The system shall allow organizers to update or delete events they manage.	Functional	High	Event Organizer	AC-EMS-F-005: Authorized changes are saved and deleted events are removed from active listings. Test: TC-EVT-02	Organizer authorization required
EMS-F-006	The system shall allow users to search and filter events using available event information.	Functional	High	User	AC-EMS-F-006: Matching events are displayed for valid search/filter criteria. Test: TC-SRCH-01	Event data required

EMS-F-007	The system shall display complete details for a selected event.	Functional	High	User	AC-EMS-F-007: Selecting an event displays its configured details and registration status. Test: TC-EVT-03	Event must exist
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4.3 Event registration and payment

Requirements for registering for events, enforcing capacity, processing applicable payments, and generating tickets.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-008	The system shall allow an authenticated user to register for an event when registration is open and capacity is available.	Functional	High	User	AC-EMS-F-008: Eligible user registration is recorded and capacity is updated. Test: TC-REG-02	Capacity and registration status required
EMS-F-009	The system shall prevent registration when an event has reached its configured capacity or registration is closed.	Functional	High	User / Organizer	AC-EMS-F-009: Registration is rejected with an appropriate message. Test: TC-REG-03	Event capacity required

EMS-F-010	The system shall process applicable registration payments through the configured payment gateway.	Functional	High	User / Payment Service	AC-EMS-F-010: Successful payment creates a confirmed payment record. Test: TC-PAY-01	Payment gateway required
EMS-F-011	The system shall generate a registration confirmation and ticket for a successfully completed registration.	Functional	High	User	AC-EMS-F-011: Confirmed registration produces a ticket containing the registration details. Test: TC-TKT-01	Successful registration required
EMS-F-012	The system shall allow users to view and cancel their eligible registrations.	Functional	Medium	User	AC-EMS-F-012: Eligible cancellation changes registration status and is reflected in the user's account. Test: TC-CAN-01	Cancellation rules required

4.4 Participant management and notifications

Requirements for organizer participant management and user notifications.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
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EMS-F-013	The system shall allow organizers to view and manage participant registrations for their events.	Functional	High	Event Organizer	AC-EMS-F-013: Organizer can view participant records only for managed events. Test: TC-PART-01	Organizer authorization required
EMS-F-014	The system shall send registration confirmations and configured event notifications to registered users.	Functional	Medium	User / Organizer	AC-EMS-F-014: Notification is generated for a configured event action. Test: TC-NOTIF-01	Notification service required

4.5 Administration

Requirements for administrative user, event, and system management.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EMS-F-015	The system shall allow administrators to manage users, approve or disable events, and review system activity.	Functional	High	Administrator	AC-EMS-F-015: Authorized admin actions are recorded and reflected in the system. Test: TC-ADMIN-01	Admin authentication required

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs are EMS-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
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EMS-NF-001	The system shall return normal user requests within 3 seconds for at least 90% of requests under normal load.	Performance	High	90th percentile response time $\leq$ 3s. Test: TC-PERF-01
EMS-NF-002	The system shall provide at least 99.5% monthly availability excluding scheduled maintenance.	Reliability	High	Monthly uptime report shows $\geq$ 99.5%. Test: TC-OPS-01
EMS-NF-003	The system shall support the expected number of concurrent users and registrations without loss of transaction data.	Scalability	High	Load test completes without data loss or critical failures. Test: TC-LOAD-01
EMS-NF-004	The system shall maintain timestamped audit logs for important user, organizer, administrator, event, registration, and payment actions.	Auditability	High	Required events are logged with timestamps and can be retrieved. Test: TC-AUD-01
EMS-NF-005	The system shall provide a responsive and understandable interface on supported desktop and mobile screen sizes.	Usability	Medium	UI evaluation passes defined responsive and usability checks. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

- Protect user, event, registration, and payment-related information from unauthorized access or disclosure.
- Ensure that only authenticated and authorized users can perform protected operations and that important security-sensitive actions are traceable.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
EMS-SR-001	All client-server communication shall use HTTPS with TLS 1.2 or higher.	Security	High	TLS configuration verified. Test: TC-SEC-01
EMS-SR-002	The system shall store user passwords only as secure password hashes and shall not store plaintext passwords.	Security	High	Database inspection shows no plaintext passwords. Test: TC-SEC-02
EMS-SR-003	The system shall enforce role-based access control for User, Event Organizer, and Administrator functions.	Security	High	Unauthorized role access is rejected. Test: TC-SEC-03
EMS-SR-004	The system shall validate and sanitize user-supplied input before processing or storing it.	Security	High	Invalid or malicious input is rejected or safely handled. Test: TC-SEC-04
EMS-SR-005	The system shall record security-relevant events such as failed logins, privileged actions, event changes, and registration/payment status changes in audit logs.	Security	High	Required security events appear in audit logs. Test: TC-SEC-05

6. Quality attributes & Acceptance tests

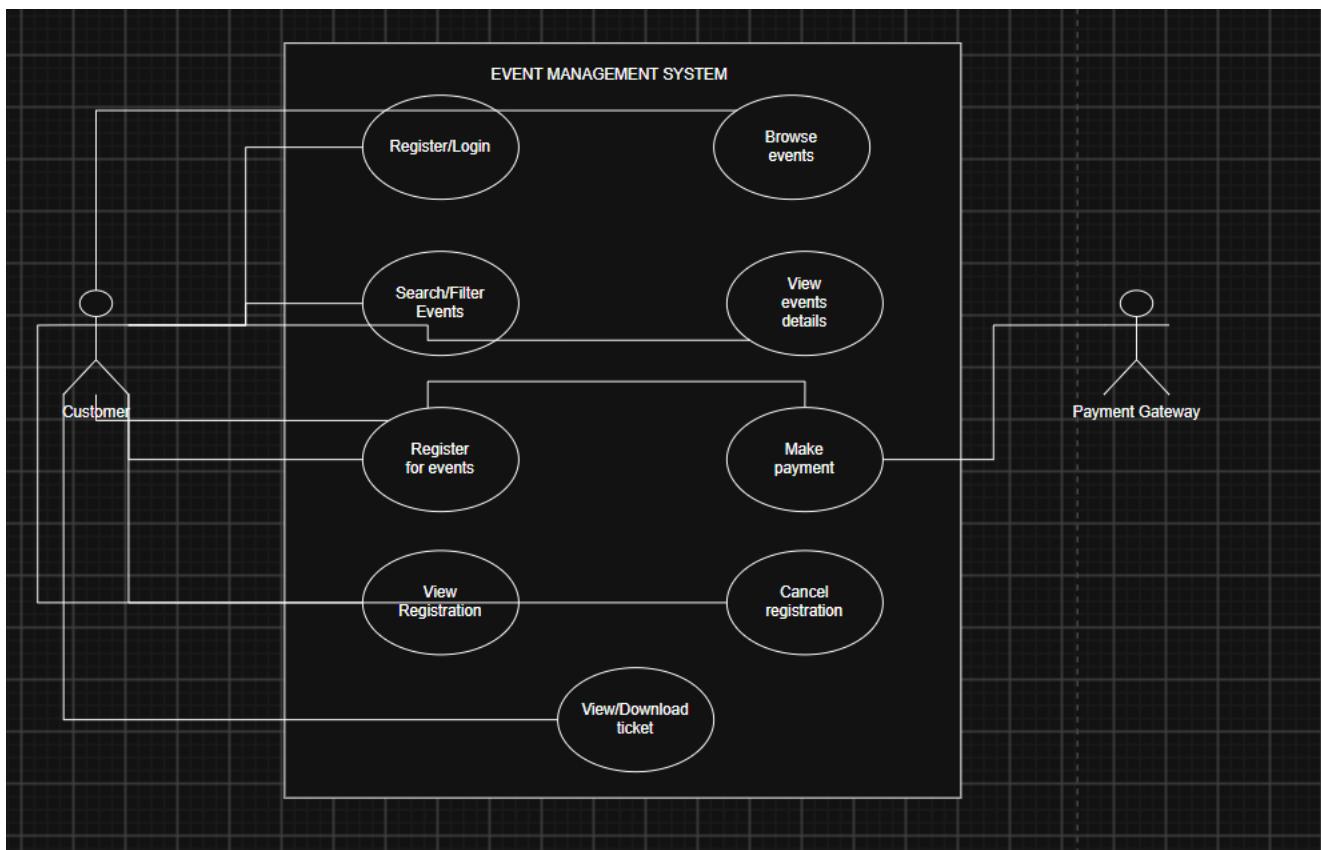
Exit criteria for acceptance: All high-priority functional requirements shall be implemented and verified, no critical non-functional or security failures shall remain, and the RTM shall show the required test cases.

Acceptance test suites: Registration and Authentication, Event Management, Search and Discovery, Event Registration, Payment, Ticketing, Participant Management, Administration, Performance, Security, Auditability, and Usability tests.

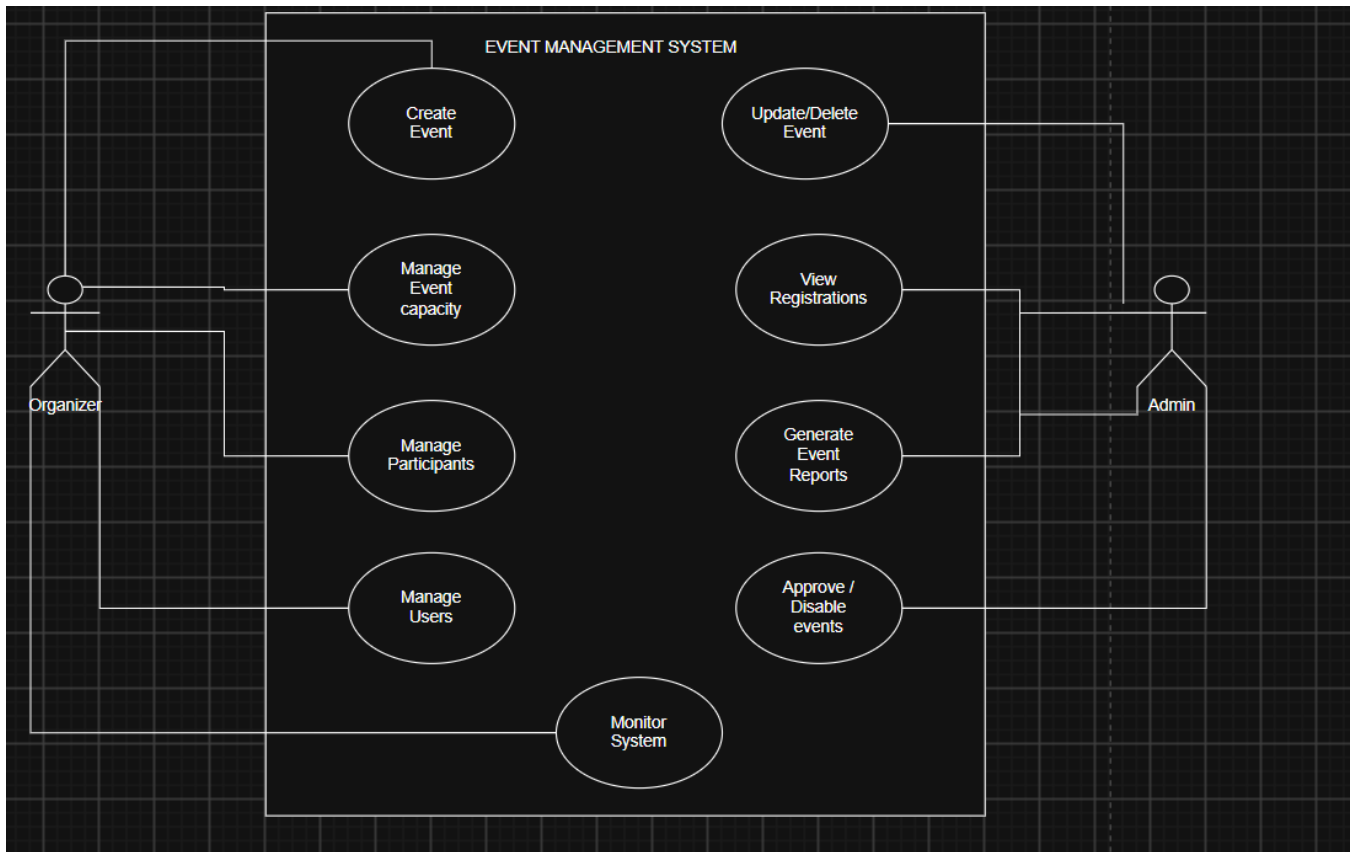
7. System models and diagrams

7.1 UML Use-Case diagram

Use-case diagram 1: Customer interactions with the Event Management System.



Use-case diagram 2: Organizer and Administrator interactions with the Event Management System.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section / Design Spec	Module	Test Case(s)	Status	Comments
EMS-F-001	Create account	4.1	User Module	TC-REG-01	N	Requires user details and database.
EMS-F-002	Authenticate user	4.1	Auth Module	TC-AUTH-01	N	Requires registered user credentials.
EMS-F-003	Update profile	4.1	User Module	TC-PROF-01	N	Requires authenticated user session.
EMS-F-004	Create event	4.2	Event Module	TC-EVT-01	N	Requires Organizer role and event data.
EMS-F-005	Update/delete event	4.2	Event Module	TC-EVT-02	N	Requires Organizer authorization.

EMS-F-006	Search/filter events	4.2	Event Module	TC-SRCH-01	N	Requires active event records.
EMS-F-007	View event details	4.2	Event Module	TC-EVT-03	N	Requires selected event record.
EMS-F-008	Register for event	4.3	Registration Module	TC-REG-02	N	Requires open registration and available capacity.
EMS-F-009	Enforce capacity	4.3	Registration Module	TC-REG-03	N	Depends on event capacity and registration status.
EMS-F-010	Process payment	4.3	Payment Module	TC-PAY-01	N	Depends on configured payment gateway.
EMS-F-011	Generate ticket	4.3	Ticket Module	TC-TKT-01	N	Depends on successful registration.
EMS-F-012	View/cancel registration	4.3	Registration Module	TC-CAN-01	N	Depends on configured cancellation rules.
EMS-F-013	Manage participants	4.4	Participant Module	TC-PART-01	N	Requires Organizer authorization and participant records.
EMS-F-014	Send notifications	4.4	Notification Module	TC-NOTIF-01	N	Depends on notification service.
EMS-F-015	Admin management	4.5	Admin Module	TC-ADMIN-01	N	Requires Administrator authentication and authorization.
EMS-NF-001	NFR: response time	5	System Module	TC-PERF-01	N	Verify using performance test under normal load.
EMS-NF-002	NFR: availability	5	System Module	TC-OPS-01	N	Verify using monthly availability monitoring.

EMS-NF-003	NFR: scalability	5	System Module	TC-LOAD-01	N	Verify using concurrent-user load testing.
EMS-NF-004	NFR: audit logging	5	System Module	TC-AUD-01	N	Verify audit records and timestamp retrieval.
EMS-NF-005	NFR: usability	5	System Module	TC-UX-01	N	Verify responsive UI on supported screen sizes.
EMS-SR-001	TLS communication	5.1.2	Security Module	TC-SEC-01	N	Verify HTTPS and TLS configuration.
EMS-SR-002	Secure password storage	5.1.2	Security Module	TC-SEC-02	N	Verify password storage in the database.
EMS-SR-003	Role-based access control	5.1.2	Security Module	TC-SEC-03	N	Verify unauthorized role access is blocked.
EMS-SR-004	Input validation	5.1.2	Security Module	TC-SEC-04	N	Verify invalid and malicious input handling.
EMS-SR-005	Security audit logging	5.1.2	Security Module	TC-SEC-05	N	Verify security events are written to audit logs.